CS5435: Text Mining & Natural Language Processing

Cross-Domain Sentiment Classification

How well do pretrained models generalize across domains?

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Problem & Motivation

The Problem

Pretrained sentiment models are widely deployed but assumed to generalize across domains without verification. Sentiment expression differs greatly between social media posts, financial news, and product reviews — vocabulary, tone, and structure all vary.

Research Questions

1. How well does a pretrained model generalize to unseen domains?

2. How much does domain shift degrade performance?

3. How much can fine-tuning recover that performance loss?

NLP Task & Datasets

Task: Multiclass Sentiment Classification (Positive · Neutral · Negative)

Twitter Airline Sentiment

Domain: Social Media (Customer Service)

Type: Short, informal tweets

Size: ~14,600 | Source: Kaggle

Class Distribution:

Neg 62.7% · Neu 21.2% · Pos 16.1%

TweetEval Sentiment

Domain: Social Media (General)

Type: Diverse tweets

Size: ~45,000 | Source: Hugging Face

Class Distribution:

Neg 15.5% · Neu 45.3% · Pos 39.1%

Financial PhraseBank

Domain: Financial News

Type: Formal sentences

Size: ~2,200 | Source: Hugging Face

Class Distribution:

Neg 13.4% · Neu 61.4% · Pos 25.2%

Methodology

Transformer Models

Twitter-RoBERTa

Pretrained on Twitter — strong social media baseline

DistilBERT

Multilingual, knowledge distillation — broad but unfocused

SentimentBERT

Pretrained on news & reviews — moderate generalization

Experimental Pipeline

① Zero-shot eval on Twitter Airline (no fine-tuning)
② Fine-tune on Twitter Airline → re-evaluate
③ Apply Airline-tuned model to TweetEval (cross-domain)
④ Fine-tune on TweetEval → re-evaluate
⑤ Apply TweetEval-tuned model to FinancialPhraseBank
⑥ Fine-tune on FinancialPhraseBank → re-evaluate

Results: Twitter Airline Dataset

Model	Pretrained Acc	Pretrained F1	Fine-tuned Acc	Fine-tuned F1
Twitter-RoBERTa	80.87%	0.7578	86.41%	0.8196
DistilBERT	19.91%	0.1466	82.27%	0.7517
SentimentBERT	47.58%	0.4556	83.95%	0.7907

Key Insight: Domain similarity matters at baseline. Twitter-RoBERTa starts at 80.87% (pretrained on Twitter data), while DistilBERT — trained on unrelated domains — starts at only 19.91%. All three models converge to 82–87% after fine-tuning, showing fine-tuning is the great equalizer.

RoBERTa
+5.54% accuracy gain

DistilBERT
+61.44% accuracy gain

SentimentBERT
+36.37% accuracy gain

Results: TweetEval Dataset

After fine-tuning on Twitter Airline, models were applied to TweetEval without retraining, then fine-tuned again.

Model	Cross-Domain Acc (no tuning)	After Fine-Tuning Acc	Macro F1 (final)
Twitter-RoBERTa	72.18%	71.95%	0.7209
DistilBERT	55.77%	65.96%	0.6542
SentimentBERT	61.09%	68.56%	0.6851

Notable Finding — RoBERTa:

Twitter-RoBERTa barely improves from 72.18% → 71.95% after fine-tuning on TweetEval. When the pretraining domain already matches the target domain well, additional fine-tuning provides little benefit and can even slightly hurt performance. In contrast, DistilBERT and SentimentBERT gain +7–10% from fine-tuning.

DistilBERT's accuracy dropped to 55.77%, indicating clear performance degradation under domain shift. The improvement after finetune is smaller, suggesting that the model had already learned some transferable representations but still required adaptation to capture broader and more diverse sentiment patterns.

Results: FinancialPhraseBank

Largest domain gap: Social media text → Formal financial sentences

Model	Cross-Domain Acc	After Fine-Tuning	Macro F1 (final)
Twitter-RoBERTa	70.48%	**97.36%**	0.9654
DistilBERT	64.90%	**96.02%**	0.9383
SentimentBERT	71.52%	**98.01%**	0.9722

Largest Gain

~26%
RoBERTa

70.48% → 97.36%

Best Final Score

98.01%
SentimentBERT

Macro F1: 0.9722

Domain Gap Effect

Severe → corrected after fine-tuning

Fine-tuning critical

Cross-Domain Analysis

① Zero-Shot Generalization Depends on Pretraining Domain

Only Twitter-RoBERTa performs well without fine-tuning because its pretraining data matches the target domain. DistilBERT (19.91%) and SentimentBERT (47.58%) fail without adaptation.

② Domain Shift Consistently Degrades Performance

Every model drops when moved to a new domain without retraining. The larger the gap (e.g., Social Media → Financial News), the greater the drop.

③ Fine-Tuning is the Essential Remedy

Across all models and all datasets, fine-tuning recovers lost accuracy. The improvement scales with the domain gap: mild shifts yield modest gains; severe gaps yield 25–60% jumps.

④ One General Model is Not Enough

No single pretrained model performs well across all three domains without adaptation, confirming that domain-specific fine-tuning is necessary for reliable deployment.

Limitations

Small Dataset

FinancialPhraseBank (~2,200 samples) may cause overfitting and overly optimistic results.

Limited Domain Coverage

Only 3 domains tested. Results may not generalize to medical, legal, or other text types.

No Hyperparameter Tuning

Models were not systematically tuned; optimal performance may be higher.

Conclusion

The results across all three phases support the hypothesis that domain shift degrades performance and that fine-tuning consistently recovers it. Domain shift degrades sentiment model performance, and the severity scales with the domain gap.

- Twitter → Twitter is a mild shift with modest performance loss.
- Twitter → Financial News is a severe shift requiring major adaptation.
- A single general-purpose pretrained model cannot reliably serve all domains.
- Lightweight domain-specific fine-tuning, even with modest data, is a highly effective remedy.

References

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